

Principles of Economics

Class 14: Economic Inequality

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Introduction

Trends in inequality

Types of inequality

Evaluating inequality

Explaining inequality

Addressing inequality

1. Introduction

- ▶ A series of observations on the level of inequality and its recent evolution.
- ▶ Source: the book The Profit Paradox.
- ▶ More at <https://www.theprofitparadox.com>

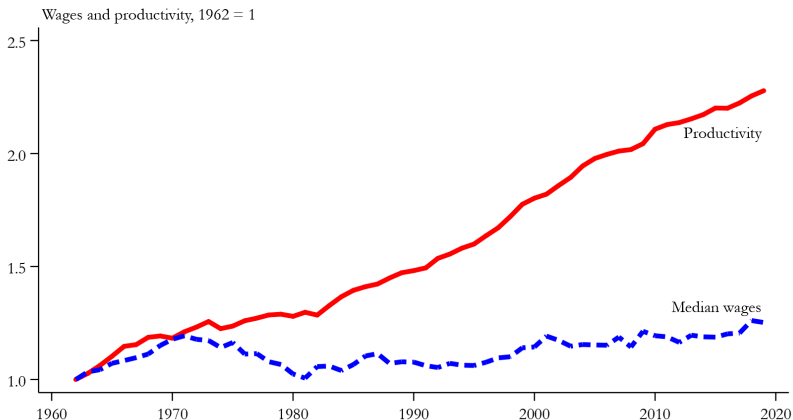


Figure IV. Median wages of all workers, and total productivity

Productivity series, see Figure 01; Wages are median wage and salary income wage of fulltime, private-sector employees in the US, excluding employees with 0 income; wages in 2012 USD (source: Current Population Survey)

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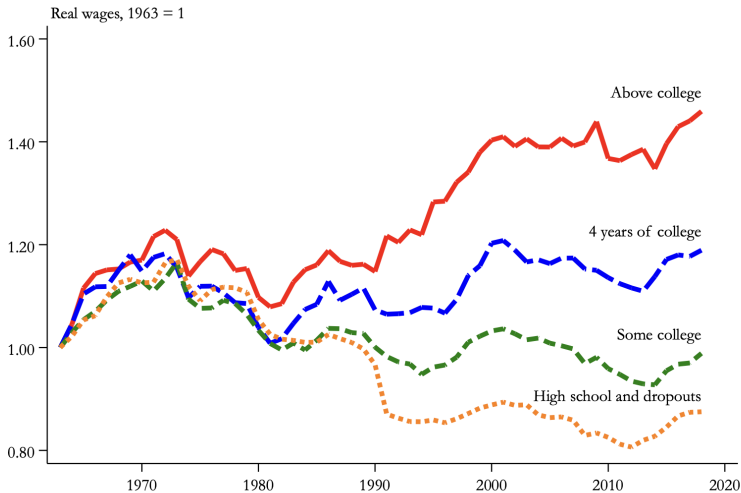


Figure 2. Wages by education

Real weekly earnings by education level (source: CPS)

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Aggregate markups

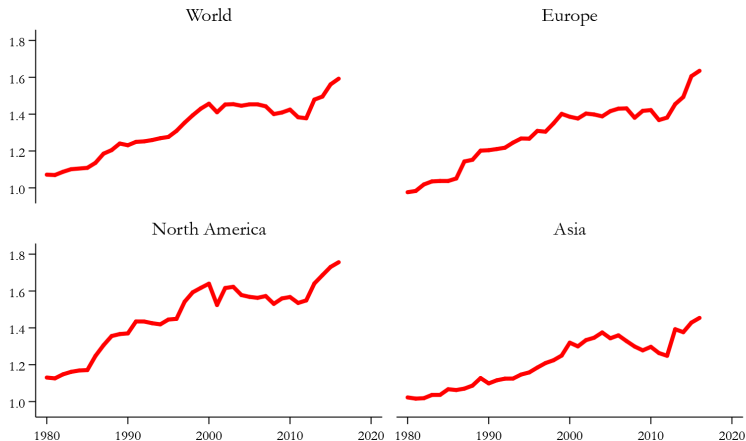


Figure 4. Aggregate global markups

Revenue-weighted average markup of publicly traded firms: World, Europe, North America and Asia (source: De Loecker & Eeckhout, 2018)

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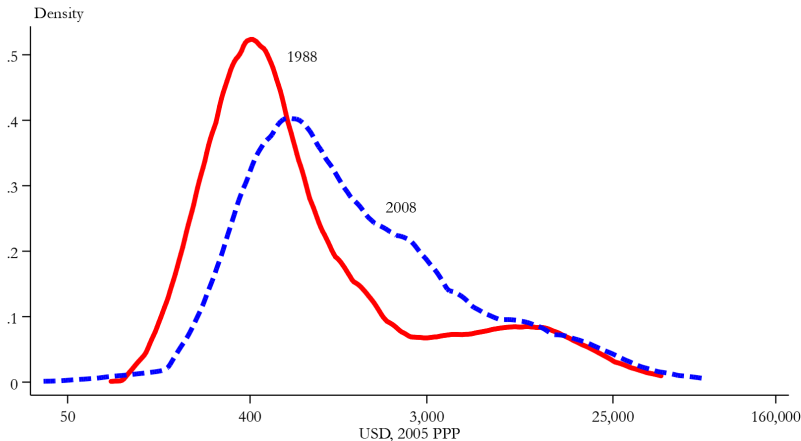
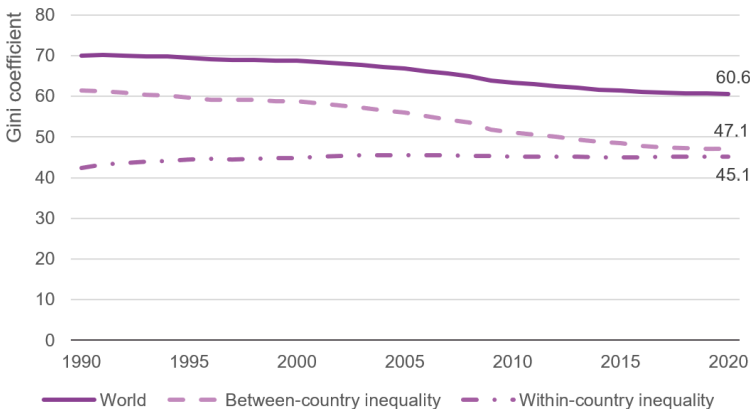


Figure XVII. World Income Distribution

Density of income (log) in the world in 1988 and 2008, in 2005 USD PPP-adjusted (source: Lakner-Milanovic (2013) World Panel Income Distribution)

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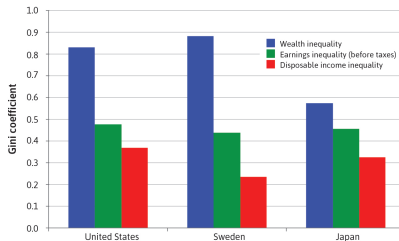


Gini coefficient: measure of inequality of a quantity (income, wealth), from zero (no inequality) to one (one individual gets everything).

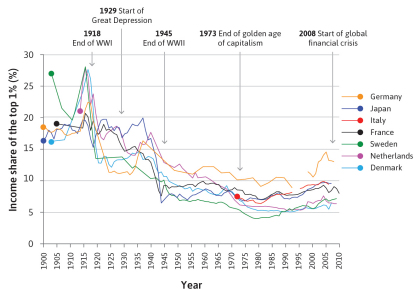
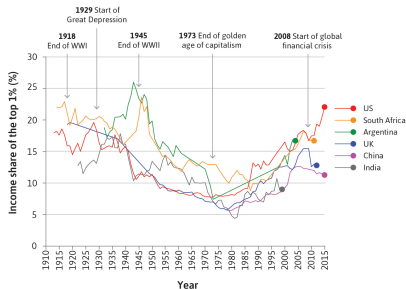
https://en.wikipedia.org/wiki/Gini_coefficient

2. Trends in inequality

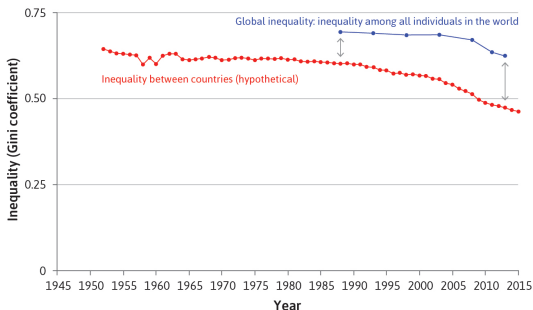
- ▶ **Market income:** Income from wages, businesses and investments
- ▶ **Disposable income:** Market income minus taxes and transfers
- ▶ Wealth is distributed more unequally than market income, and market income more unequally than disposable income.
- ▶ Likely cause of the greater equality in disposable income: taxes and transfers.



- ▶ Countries differ in the level of inequality, but share trends: it fell between 1920 and 1980.
- ▶ After 1980 the paths diverge.



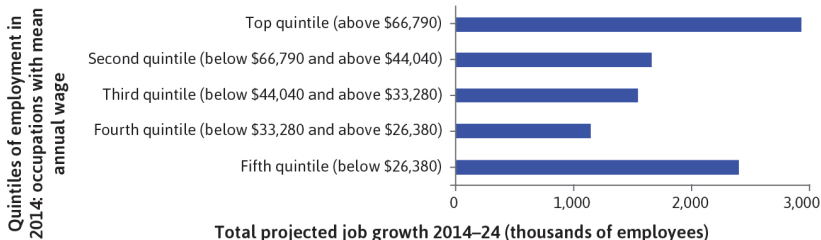
- ▶ Since 1980: inequality between countries falls fast, inequality within each country rises.
- ▶ Net result: global inequality has begun to fall.



Blue = Gini for the whole world.

Red = if everyone in each country earned the average income.

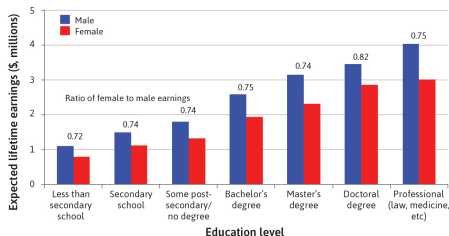
- ▶ Rising inequality in most countries goes with a changing distribution of jobs.



- ▶ “Missing middle”: more low-wage and high-wage jobs, fewer middle-wage jobs.

3. Types of inequality

- ▶ Categorical inequality (group inequality): economic differences between people treated as different categories.
- ▶ Usually based on “accidents of birth”:
 - ▶ Country of citizenship: passports and borders limit economic opportunities
 - ▶ Gender or ethnic group
- ▶ Example: income gap between men and women with the same education.



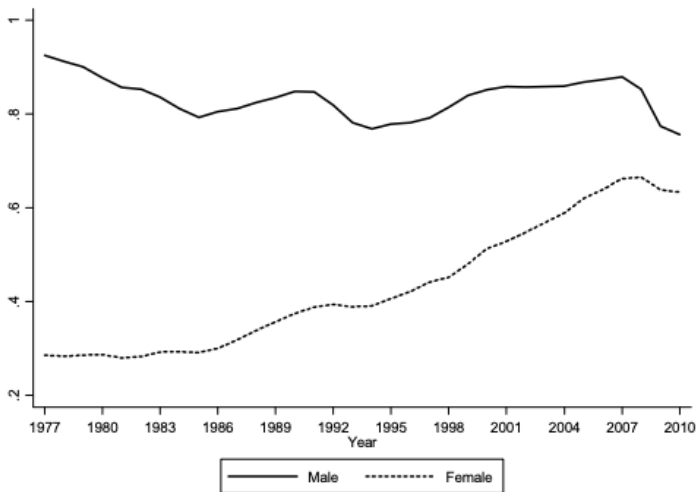
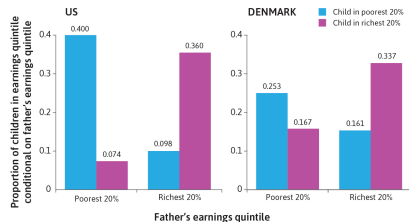
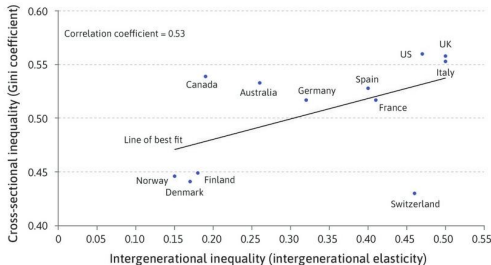


Figure 5. Employment rate

- ▶ **Intergenerational inequality:** How much of the parents' differences passes to the next generation.
- ▶ The process of **intergenerational transmission** takes many forms:
 - ▶ Inheriting parents' wealth
 - ▶ Inheriting parents' genetics
 - ▶ Parents' influence on education
- ▶ **Intergenerational elasticity:** percentage difference in the second generation's status per 1% difference in the adult generation's. (High elasticity = low **intergenerational mobility**)

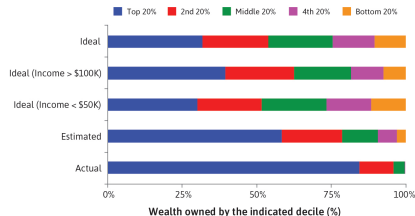


- ▶ Income inequality correlates positively with intergenerational inequality.
- ▶ Possible reasons:
 - ▶ A strong culture of fairness brings policies that cut cross-sectional inequality and raise intergenerational mobility.
 - ▶ Good and bad shocks (“luck”) pass to the next generation and feed cross-sectional inequality.

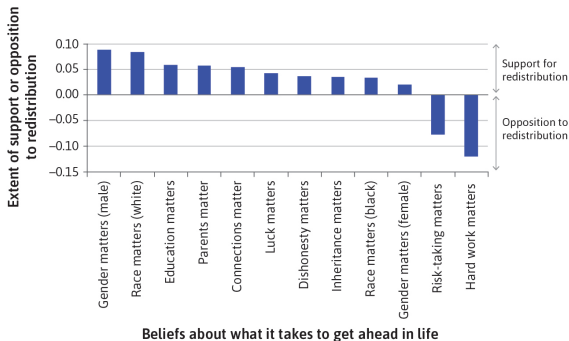


4. Evaluating inequality

- ▶ Inequality is a problem if there is “too much”.
- ▶ Example (US): actual wealth inequality is greater than what people estimate, and that estimate already exceeds what they see as ideal.
- ▶ The ideal distribution varies with income, but far less than that gap.

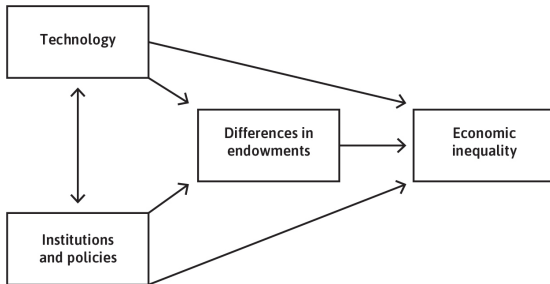


- ▶ Whether inequality is seen as unfair depends on beliefs about its origin.
- ▶ Many do not see inequality from effort or risk-taking as unfair.

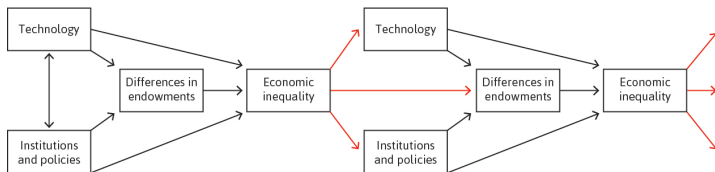


5. Explaining inequality

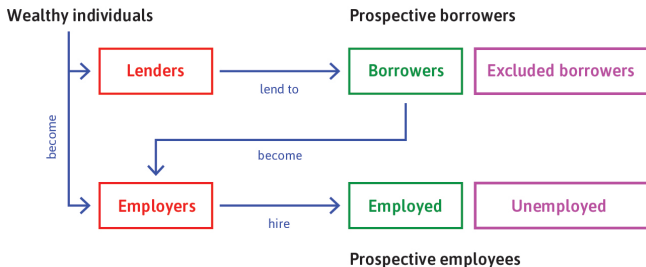
- An individual's income depends on:
1. Their **endowments**: what they have that affects income (wealth, physical assets, **human capital**).
 2. The value of each item in the endowment, set by technology and institutions.



- ▶ Economic inequality comes from institutions, technology and differences in endowments.
- ▶ It also feeds back on future institutions and technology: political advantage for the rich; automation triggered by the minimum wage.



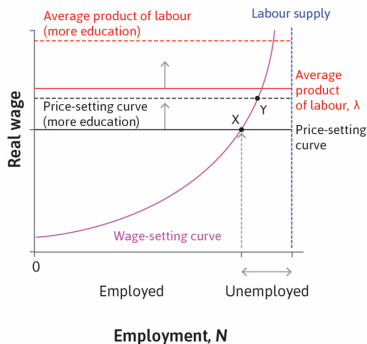
- Differences in endowments shift the balance of power: in **principal-agent** relations the principal has power over the agent, not the reverse.



- They also decide who becomes the principal and who the agent: who can lend and who must borrow.

- A better educated labour force raises productivity: for example, a higher compulsory schooling age.

At the initial wage, firms make profits.
↓
More firms enter, reducing unemployment.
↓
Better reservation option, so higher wages.
↓
Inequality falls.



- ▶ Primary labour market: “good” jobs, high wages, job security, unions.
- ▶ Secondary labour market: short contracts, low wages, little job security.
- ▶ Removing segmentation raises average wages, reducing inequality.

- ▶ **Automation:** Machines do work that people used to do.
- ▶ New technologies raise the demand for some skills and cut it for others.
- ▶ In the short run:
 - ▶ Machines replace routine work, increasing unemployment.
 - ▶ Workers with complementary skills earn more: inequality rises.

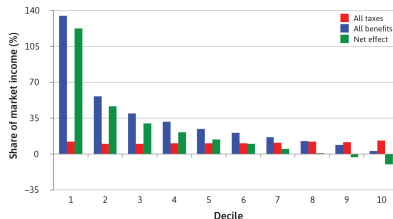
- ▶ In the long run, two opposing effects on wages and employment:
 1. More unemployment worsens the reservation option and lowers the wage firms must set.
 2. But higher productivity raises profits, which finance new capital, new jobs and less unemployment: the wage rises.
- ▶ Policies can ease the transition:
 - ▶ Opportunities for displaced workers to improve their skills.
 - ▶ Job opportunities and higher wages in non-routine sectors.

6. Addressing inequality

- ▶ Public policy can act on economic inequality:
 - ▶ **Redistribution**: Taxes and transfers that cut differences in disposable income and in public services.
 - ▶ **Predistribution** - More equal endowments: redistributing property, or raising by law the value of the endowments of the poorest.
- ▶ Example: a **legal minimum wage** for specific types of workers.

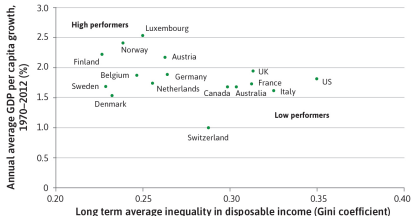
Endowment	Policy	Direct effect	Indirect effect	Unit
Labour	Free high-quality education for all children.	Increases the opportunities of the poorest children to reach higher levels of education, which raises the market value of their labour endowment.	Raises the average productivity of labour, shifting the price-setting curve up, which raises wages and employment (<i>ceteris paribus</i>).	19
Labour	Increase in the share of the harvest retained by the farmer.	Increases the value of the labour endowment of farmers.	Increases the incomes of farmers.	5
Labour	Removal of ethnic, racial or gender discrimination.	Increases the value of the labour endowment of those subject to this discrimination.	Increases the incomes of these groups.	19
Labour	Minimum wage	Increases the value of the labour endowment of those working at the minimum wage.	Increases the incomes of the poor and reduces those of employers (unless employment effects dominate).	19
Labour	Laws to increase workers' bargaining power (unions).	Increases the value of the endowment and the working conditions of union members.	Increases the incomes of union members (unless employment and productivity effects dominate) and reduces those of employers.	9, 16, 19
Ownership of firms	Measures to ensure competition.	Reduces profit margins.	Raises real wages and reduces profits.	7, 9, 16
Intellectual property	Restricting intellectual property rights (shorter patents or copyrights).	Reduces the value of the intellectual property endowment.	May discourage innovation, but speeds up its diffusion.	21
Professional licence	Making access to licences easier (taxis).	Raises supply and reduces the incomes of licence holders.	Greater equality (if licence holders are richer than average).	19

- ▶ **Welfare state:** policies that convert market income into final income.
- ▶ Two legs: 1) taxes and 2) spending, with **in-kind transfers** (free or subsidised services) and **social insurance** (for specific groups).
- ▶ **Progressive** policies directly reduce inequality
- ▶ **Regressive** policies directly increase inequality.
- ▶ Example: in Mexico, spending is more progressive than taxes.

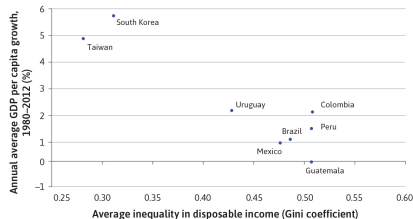


Trends	Causes
Falling internal inequality (1920-1980)	a) Improvements in education and productivity reduced unemployment. b) Labour market segmentation and other causes of inequality between workers fell (including c, below). c) Technological improvements that were complementary to the work of low and medium skilled employees.
Stable or rising internal inequality (1980-2017)	a) Rising inequality between workers due to new technologies: i. Complementary to the work of high skilled employees. ii. Substituting for routine jobs. b) Weaker unions and a shift of bargaining power towards employers, while the resulting rise in after-tax profits did not translate into more employment (in some countries).
Stable or falling external inequality (1995-2017)	Reduction in global labour market segmentation due to rapid growth in labour productivity and demand in China and other developing countries.

Rich countries

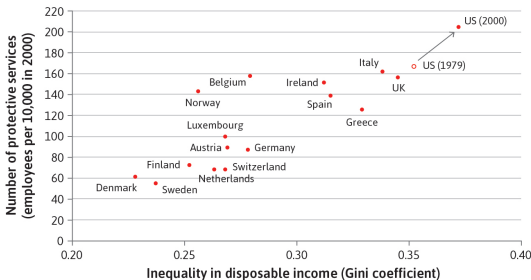


Emerging countries



- Growth rates are similar and uncorrelated with inequality. High taxes and transfers need not blunt the incentives to work or innovate.

- ▶ Economic benefits of lower inequality:
 - ▶ High inequality makes the trust and cooperation behind knowledge and care services harder to sustain.
 - ▶ Policies that improve the endowments of the poor improve productivity.
 - ▶ Less labour tied up in security, which drains other sectors



What you take home from each section:

1. **Introduction** → inequality is large and has changed shape: median wages lag productivity and aggregate markups keep rising.
2. **Trends in inequality** → it fell between 1920 and 1980 and the paths diverged after 1980; between countries it falls, within each country it rises, and the middle of the job distribution empties out.
3. **Types of inequality** → categorical inequality comes from accidents of birth, and intergenerational elasticity measures how much is inherited: more unequal countries are also less mobile.
4. **Evaluating inequality** → how much is too much is a judgement: people underestimate actual inequality, and they call it unfair only when they think it does not come from effort or risk.
5. **Explaining inequality** → income is endowments times their value; technology and institutions set that value and the balance of power, so automation and labour market segmentation move inequality.
6. **Addressing inequality** → redistribution acts after the market and predistribution before it; growth is not lower where redistribution is larger, and less inequality has economic benefits of its own.

Key concepts: Gini coefficient, market income, disposable income, categorical inequality, intergenerational transmission, intergenerational elasticity, intergenerational mobility, endowments, human capital, principal-agent relations, primary and secondary labour market, automation, redistribution, predistribution, welfare state, in-kind transfers, social insurance, progressive and regressive policies.